

Personalized Risk-to-Timeline Modelling for Cervical Cancer Progression

Predicting CIN outcomes from clinical and population risk factors

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SAT 5141 – Clinical Decision Support & AI Modelling

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Population & Problem Justification

Target population

- Women at risk of cervical intraepithelial neoplasia (CIN) and cervical cancer
- High burden in low- and middle-income countries due to late detection and limited follow-up

Why this population?

- Cervical cancer is preventable but still a leading cause of cancer death among women
- Persistent high-risk HPV infection, smoking, reproductive and sexual factors drive progression
- CIN exists along a progression timeline – ideal for Risk-to-Timeline (R-to-T) modelling

Clinical problem

- Most ML work focuses on static “disease present vs not” classification
- Clinicians need tools that estimate individual risk and help plan follow-up timelines

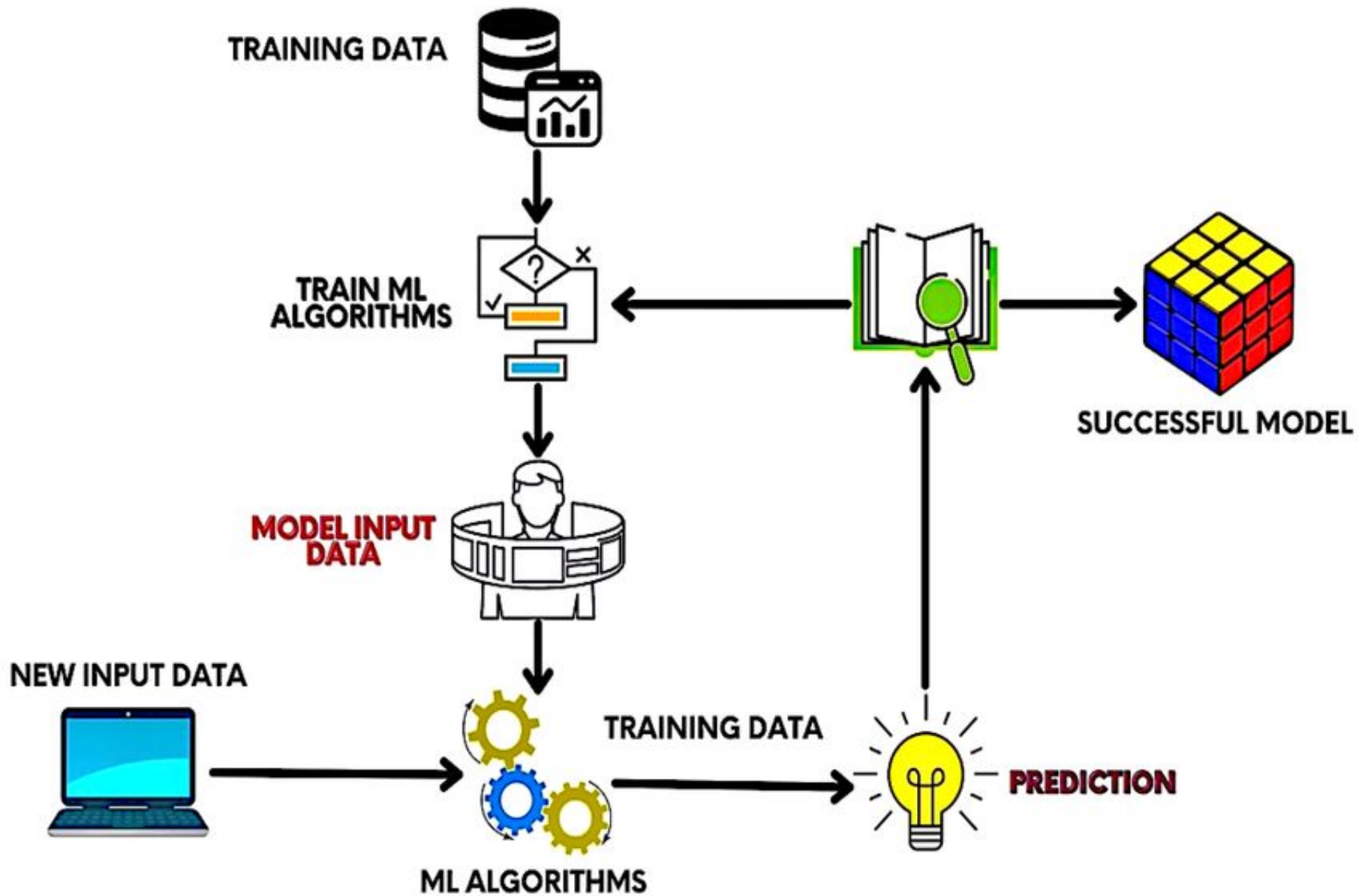
Literature Review & Motivation

Evidence base for AI in cervical cancer (≥ 7 primary studies)

- Ahishakiye et al. 2020 – Ensemble learning on risk factors for cervical cancer prediction
- Dellino et al. 2024 – AI-enhanced screening and personalized risk profiles for HPV-positive women
- Kondo, Ngondya & Rusheke 2025 – Systematic review of UCI cervical cancer ML studies and limitations
- Mahto & Sood 2024 – Handling severe imbalance with ADASYN and tuned ensembles on small risk datasets
- Rothberg et al. 2018 – Hierarchical regression on EHR data to stratify CIN2+ risk
- Shakil et al. 2024 – ML + feature selection + explainable AI for precise cervical cancer detection
- Yang et al. 2019 – ML model for cervical cancer risk prediction and factor analysis

Key gaps motivating this project

- Heavy reliance on the UCI dataset but inconsistent preprocessing and imbalance handling
- Most work predicts presence of disease rather than modelling progression over time
- Limited focus on interpretability and clinical usability of models



Almudawi et al. (2022). A Model for Predicting Cervical Cancer Using Machine Learning Algorithms.

Dataset, Frequency & Cost Burden

Dataset characteristics

- UCI Cervical Cancer Risk Factors dataset – 858 women, 36 attributes
- Demographic, behavioural, reproductive and STD history
- Diagnostic tests: Hinselmann, Schiller, Cytology, Biopsy

Outcome of interest

- Dx:CIN (biopsy-confirmed cervical intraepithelial neoplasia)
- Only 9 CIN-positive cases ($\approx 1\%$) → extreme class imbalance

Burden and costs (context from literature)

- Cervical cancer causes major morbidity and mortality globally despite vaccination and screening
- Screening pathways (Pap smear, HPV testing, colposcopy, biopsy) are resource-intensive
- Delays in detecting CIN and progression to cancer increase treatment costs and loss of life-years

Project aim

- Use low-cost risk-factor data + diagnostic results to prioritize high-risk women for timely follow-up

Clinical Measurements, Diagnoses & Codes

- Key diagnostic variables used as measurements
 - Dx:CIN – biopsy-confirmed CIN (primary target label)
 - Dx:Cancer, Dx:HPV, Dx – additional diagnostic flags
 - Hinselmann, Schiller, Cytology, Biopsy – clinical test outcomes feeding risk
- Risk factor measurements
 - Age, number of sexual partners, age at first intercourse, number of pregnancies
 - Smoking status, years, packs/year
 - Hormonal contraceptive use and duration; IUD use and duration
 - STD history: type, number of diagnoses, including HPV, HIV, syphilis, etc.
- Mapping to the “codes” perspective
 - Each Dx and test variable corresponds to a diagnostic event, analogous to diagnosis/procedure codes
 - Drug-related exposure is proxied by contraception and smoking variables

Measurement Strategy & Algorithms

- Primary measurement
 - Binary classification task: predict Dx:CIN (0 = no CIN, 1 = CIN present)
 - Predictors: combined demographic, behavioural, STD and diagnostic factors
- Algorithms implemented
 - Baseline and advanced models:
 - K-Nearest Neighbors (KNN)
 - Gaussian Naive Bayes
 - Decision Tree (regularized)
 - Multi-Layer Perceptron (MLP)
 - Support Vector Machines – Linear, RBF, Sigmoid kernels
 - Gradient Boosting (regularized)
 - Easy Ensemble (ensemble for imbalance)
- Why focus on Linear SVM
 - Showed the best balance of recall and precision for CIN-positive cases under 3-fold stratified CV
 - Achieved high minority-class F1 while maintaining very high overall accuracy
 - Stable across folds despite very small number of positives

Alternative Tools & Modelling Options

Algorithms we
could have used
(but did not in
this phase)



Random Forests and XGBoost for non-linear interactions and better handling of imbalance

Logistic Regression as a transparent baseline for odds interpretation

Survival models (Cox Proportional Hazards, survival trees) for true time-to-event prediction

Deep learning models for image-based or multi-modal integration if colposcopy images were available

Additional tools
and extensions

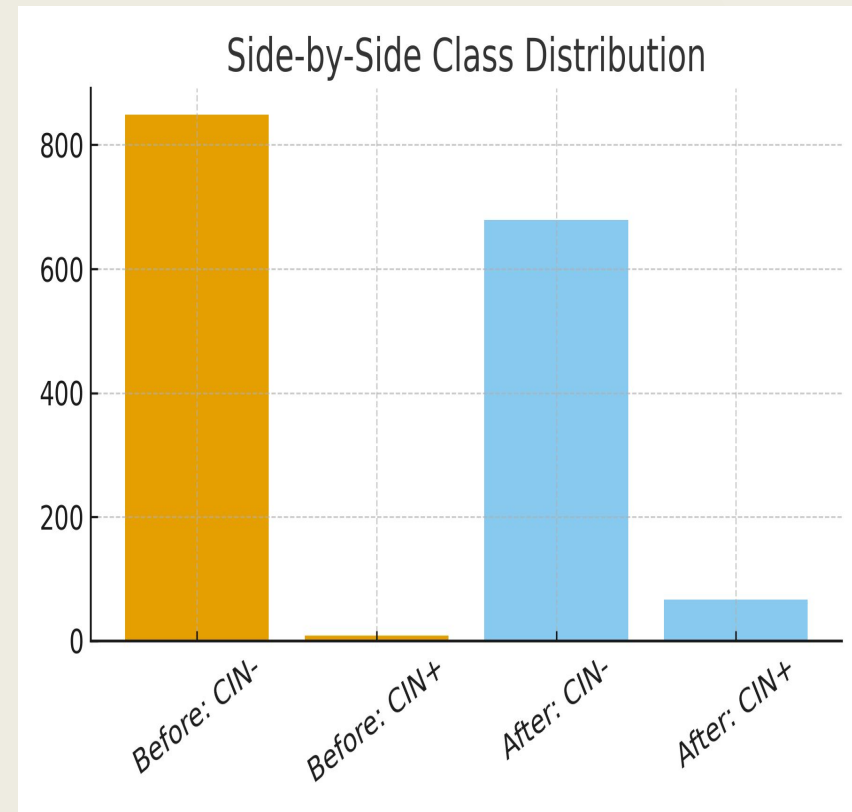
calibration plots to assess probability estimates

Cost-sensitive learning to weight false negatives more heavily than false positives

Real-time EHR integration for prospective decision support

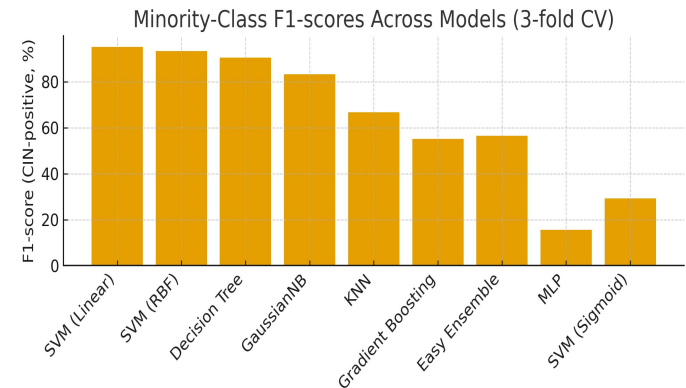
Data Strengths, Weaknesses & Risk Adjustment

- Strengths of the dataset
 - Contains rich mix of behavioural, demographic and diagnostic variables in one table
 - Publicly available and widely used → enables comparison with prior work
 - Includes multiple test modalities (Hinselmann, Schiller, Cytology, Biopsy)
- Key weaknesses and limitations
 - Extreme imbalance: only 9 CIN-positive out of 858 records
 - Substantial missing data; two STD-time variables >90% missing and dropped
 - Many numeric fields stored as text, requiring coercion and cleaning
 - No temporal information – a “snapshot” dataset (no true timelines)
- Data cleaning and risk adjustment
 - Replaced “?” with NaN and performed median imputation for numeric fields
 - Dropped highly sparse columns (STD time-since variables)
 - Stratified 80/20 train–test split to preserve class proportions
 - SMOTE applied to training data to create 10% CIN-positive prevalence
 - Class weights computed and passed into applicable models to penalize false negatives



Model Evaluation & Key Results

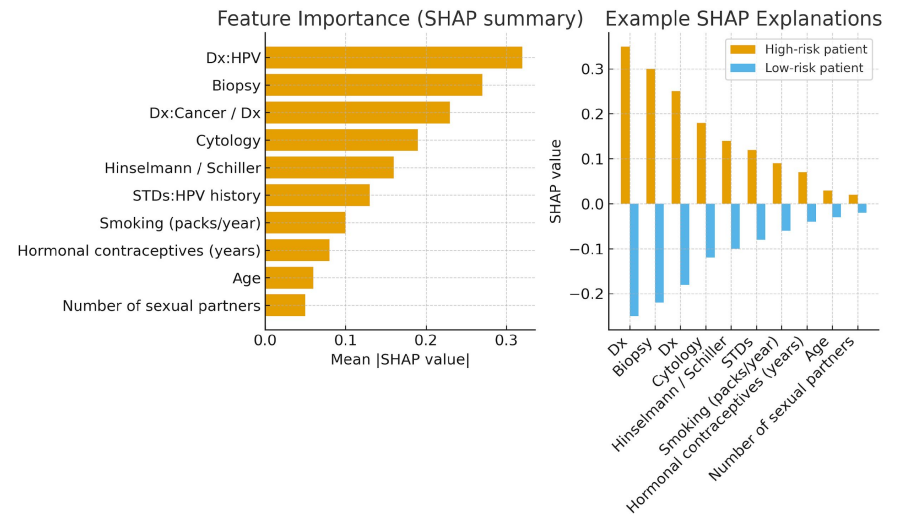
- Evaluation approach
 - Initial single 80/20 train–test split on original data
 - Weighted metrics: Accuracy, Precision, Recall, F1-score
 - Limitations: only 2 CIN-positive cases in test set → unstable estimates
 - More robust 3-fold stratified cross-validation
 - Performed on original dataset without SMOTE-generated samples
 - Provided average metrics across folds for CIN-positive class
- Headline results (3-fold CV)
 - Linear SVM: best balance of recall and precision; highest F1 for CIN-positive cases
 - RBF SVM: high precision but lower recall
 - Decision Tree: perfect recall but lower precision
 - Easy Ensemble: very high recall but poor precision → many false positives
 - MLP and Sigmoid SVM: struggled to consistently detect positives



Model	Accuracy	Precision	Recall	F1-score
SVM (Linear)	99.9	91.7	100.0	95.2
SVM (RBF)	99.9	100.0	88.9	93.3
Decision Tree	99.8	83.3	100.0	90.5
GaussianNB	99.8	100.0	77.8	83.3
KNN	99.5	100.0	55.6	66.7
Gradient Boosting	99.4	58.3	55.6	55.2
Easy Ensemble	98.3	40.0	100.0	56.5
MLP	96.5	10.4	33.3	15.6
SVM (Sigmoid)	90.8	19.3	100.0	29.3

Explainability with SHAP (Linear SVM)

- Why interpretability matters
 - Clinicians must understand why a model flags a woman as high-risk
 - Explainable AI builds trust and supports shared decision-making
- SHAP analysis for Linear SVM
 - Used KernelExplainer to compute SHAP values for the best-performing SVM
 - Generated global summary plot to rank features by importance
 - Local explanation examples show how individual risk factors shift CIN predictions
- Key risk factors identified
 - HPV-related variables and prior abnormal Dx (Dx:HPV, Dx:Cancer, Dx)
 - Abnormal diagnostic tests (Hinselmann, Schiller, Cytology, Biopsy)
 - Behavioural factors such as smoking and STD history



Interpretation of Findings & Population Insights

What we learned about the population

- Most women in this dataset are low-risk for biopsy-confirmed CIN ($\approx 99\%$ negative)
- A small subset shows strong risk patterns driven by HPV and abnormal tests
- Behavioural factors (e.g., smoking, contraceptives, STDs) align with known clinical risk pathways

Model behaviour under imbalance

- High overall accuracy can be misleading when positives are rare
- Cross-validation revealed which models truly detect CIN vs which only learn the majority class
- Linear SVM provided the best trade-off between sensitivity and specificity for CIN

Clinical meaning

- The model can help identify a small minority of women who need urgent follow-up
- Explainability confirms that predictions are driven by clinically plausible factors

Recommendations to Improve the Measure

Data-related recommendations

- Collect larger, multi-site datasets with more CIN-positive and CIN2+ cases
- Include true temporal information: dates of tests, diagnoses and follow-up visits
- Link EHR data (labs, procedures, medications) and, where possible, colposcopy images

Modelling recommendations

- Extend from static classification to survival or time-to-event models (Cox, survival GBMs)
- Evaluate advanced ensembles (XGBoost, Random Forest) with calibrated probabilities
- Incorporate cost-sensitive learning to explicitly penalize missed CIN cases

Implementation recommendations

- Embed the model into a clinical decision support system as a “risk-flagging” tool
- Use XAI-based dashboards so clinicians can see key drivers for each prediction

Feasibility of Recommendations

Short-term feasibility

- Re-training with larger, better-curated versions of the UCI dataset is technically easy
- Adding Random Forest/XGBoost, improved cross-validation and calibration is straightforward
- Explainability tools (e.g., SHAP) are already integrated into the workflow

Medium-term feasibility

- Obtaining access to real hospital EHR data requires ethics approval and data-sharing agreements
- Survival modelling is feasible once longitudinal data become available
- Integration into existing screening workflows needs stakeholder engagement

Long-term considerations

- Deployment in low-resource settings requires simple interfaces and local validation
- Robust external validation is needed before routine clinical use

Overall assessment

- Methodological enhancements are feasible now; full risk-to-timeline clinical deployment will require multi-year, multi-institutional collaboration

Conclusion

- We developed and compared multiple ML models on the UCI cervical cancer risk dataset
- After addressing imbalance and using stratified cross-validation, Linear SVM emerged as the most reliable CIN predictor
- SHAP analysis confirmed that the model relies on clinically meaningful risk factors
- This work establishes a solid predictive foundation for future Risk-to-Timeline cervical cancer progression modelling

References

- Ahishakiye et al. 2020 – Ensemble learning for cervical cancer risk
- Shakil et al. 2024 – Explainable ML for cervical cancer detection
- Yang et al. 2019 – Risk prediction model using ML
- Rothberg et al. 2018 – EHR-based CIN2+ risk stratification
- Kondo et al. 2025 – Systematic review of cervical cancer ML
- Mahto & Sood 2024 – Imbalance handling with ADASYN
- Dellino et al. 2024 – AI in cervical cancer screening